Hyperagents: agents that edit their own improver

Jenny Zhang, Bingchen Zhao, Wannan Yang, Jakob Foerster, Jeff Clune, Minqi Jiang, Sam Devlin, Tatiana Shavrina · UBC · Vector Institute · Edinburgh · NYU · Meta · arXiv 2603.19461v1

HarnessRSI 74 ★★★

evidence: improver

Paper

Code

1 What changes, what is kept

Hyperagent = ONE editable program

Task agent
solves the task

Meta agent
edits agents (incl. itself)

DGM: the instruction-generating step is fixed, handcrafted

1 Archive of hyperagents
starts from one simple agent; every child that compiles is added

2 Sample a parent
sigmoid(score vs top-3 mean) × 1/(1 + compiled children)

3 Meta agent edits the repo
sees code, past results, iterations left; may rewrite itself

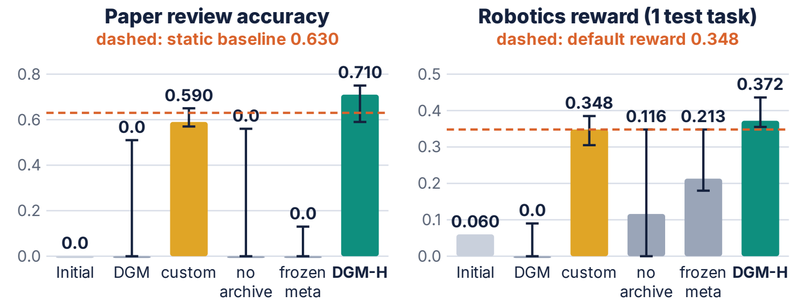
4 Evaluate the child
10 training tasks first, full set only if it passes; val score if any

5 Add to archive if it compiles
no accept/reject by score; selection happens at step 2

100×

end: best by validation → one test evaluation
fixed, not self-modifiable: parent selection, evaluation protocol, task distribution

2 Headline: test tasks, 5 runs per method



Medians, whiskers = 95% CI. Gold = hand-customised DGM; gray = ablations. Both domains optimised jointly in one run. Initial agent: 0.0 on paper review = fails the output format.

4 Where the gain is absent or thin

- **vs hand-customised DGM: not significant** ($p > 0.05$), CIs overlap; robotics is one test task (0.372 vs default 0.348)
- Paper review: vs static baseline 0.710 vs 0.630, not vs 0.0; CI 0.590–0.750 contains 0.630; baseline has no variance
- Coding (80 iterations): 0.267 vs the DGM paper's reported 0.307 (starts 0.084 vs 0.142)
- Compounding n.s.; 0.700 vs 0.670 = one agent, 3 tasks, no CI; 0.561 → 0.601 includes optimisation tasks
- Self-modifiable parent selection 0.491 vs handcrafted 0.532 (random 0.428; both-domain avg), n.s.

3 Controls that decide if the gain is real

Meta-level ablation · same 100 iterations

test median (95% CI)	paper review	robotics
DGM-H	**0.710 (0.590–0.750)**	**0.372 (0.355–0.436)**
DGM-custom **p>0.05**	0.590 (0.570–0.650)	0.348 (0.305–0.385)
frozen meta **p<0.05**	0.0 (0.0–0.130)	0.213 (0.180–0.348)

No archive: 0.0 / 0.116; DGM 0.0 / 0.0. n = 5 runs; p-values as reported, test details unspecified. Baseline token cost not reported.

Does the improver transfer? Math grading, 5 runs

start of a 50-iteration run	imp@50 (CI)
initial meta agent	0.0 (0.0–0.130)
meta agent from DGM-custom runs	0.0 (0.0–0.010)
hyperagent from DGM-H runs	**0.630 (0.540–0.630)**

Arms differ in source runs and generation procedure; no split of format repair vs grading skill in 0.630. $p<0.05$ as reported, n = 5.

Does it compound?

200 iterations, math grading	test (CI)
DGM-H from initial agent	0.610 (0.510–0.680)
DGM-H + transferred hyperagent **p>0.05**	0.640 (0.550–0.720)

Cost of DGM-H

≈ 33M tokens self-modification + 0.506M per iteration evaluation. Paper states ≈ 88.6M per 100-iteration run; its parts sum to 83.6M‡.

Takeaway

Letting the meta agent edit itself beats a frozen meta agent, a no-archive search and the coding-tuned DGM, and the learned improver carries over to a new domain. Against a hand-customised improver the gap is not significant, and the evidence rests on 5 runs per method and a few test sets.

Our read · HarnessRSI

Score 74 · ★★★ · evidence level: improver

Reports: test splits, 5 runs + CIs, ablations, transfer, cost.
Does not report: baseline token cost, leakage audit.